Léo Delalandre
PhD

Curriculum Vitae
March 2026

Location : Complexity Science Hub, Vienna, Austria

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Career summary

2026-2028 **Postdoc**, *Complexity Science Hub*, *Vienna*, *Austria*, co-supervised by Alex Mesoudi and Peter Turchin. "The cultural evolution of gender inequalities".

2020-2024 **Ph.D. in Functional Ecology**, *Centre for Evolutionary and Functional Ecology (CEFE)*, *Montpellier*, *France*, supervised by Cyrille Violle and Eric Garnier. "Trait-environment relationships in plants: from organism life cycle to data life cycle".

2020 **Research intern,** *Centre for Evolutionary and Functional Ecology (CEFE)*, *Montpellier*, *France*, supervised by Cyrille Violle. Modelling the effects of original tree species on forest productivity.

2017 **Research intern**, *Centre for Investigations on Desertification (CIDE)*, *Valencia*, *Spain*, supervised by Alicia Montenisos-Navarro and Miguel Verdú. Inferring facilitation between plants from co-occurrence patterns.

2016 **Research intern,** *Institute of Ecology and Environmental Sciences (IEES)*, *Paris*, *France*, supervised by Elisa Thébault and Colin Fontaine. Modelling the effects of phenology on plant-pollinator interactions.

Education

2020-2024	**Ph.D. in Functional Ecology** *Montpellier, France*	École Doctorale 584 GAIA, Univ. Montpellier
2019-2020	**Master 2: Biodiversity, Ecology, Evolution** *Paris, France*	Université Paris Sud, Muséum National d'Histoire Naturelle
2018-2019	**Master 2**: *Agrégation* (teaching concours) in Biology and Geology *Lyon, France*	ENS (École Normale Supérieure) of Lyon
2016-2017	**Master 1: Biosciences** *Lyon, France*	ENS (École Normale Supérieure) of Lyon Univ. Claude Bernard Lyon 1
2015-2016	**License: Biosciences** *Lyon, France*	ENS (École Normale Supérieure) of Lyon Univ. Claude Bernard Lyon 1
2013-2015	**Preparatory school: Biology, Chemistry, Physics, Earth Sciences** *Rouen, France*	Lycée Pierre Corneille

Peer-reviewed publications

In preparation

In revision

Delalandre, L., Segrestin, J., Barkaoui, K., Violle, C., and Garnier, E. (2026). From FAIR to MEGaFAIR: a checklist to future-proof trait data in ecology and evolution. Submitted to *Ecology* in March 2026.

Violle, C., Mahaut, L. , Mouillot, D., Díaz, S., Mouquet, N., Gauzere, P., Enquist, B.J., Maire, A., Grenié, M., O'Connor, Louise, Kraft, N.J.B, Auber, A., Cadotte, M.W., Cardou, F., **Delalandre, L.**, Judah, A.B., Klausmeier, C.A., Litchman, E., Loiseau, N., Maitner, B., McGill, B.J., McLean, Matthew, Munoz, F., Ostling, A., Pimiento Hernandez, C., Toussaint, A., Villéger, S., Zinger, L., Thuiller, W. Trait-based pathways to protect nature and its contributions to people. In major revision for PNAS (January 2026).

Published

Delalandre, L., Violle, C., Fort, F., Tschambser, J., Saugier, L., Fourtier, G., & Garnier, E. (2025). Plant response to nutrients differs among traits and depends on species' nutrient requirements. *Annals of Botany*, mcaf171. https://doi.org/10.1093/aob/mcaf171

Garnier, É., **Delalandre, L.**, Segrestin, J., Barkaoui, K., Kazakou, E., Navas, M.-L., Vile, D., Violle, C., Bernard-Verdier, M., Birouste, M., Blanchard, A., Bumb, I., Cruz, P., Debain, S., Fayolle, A., Fortunel, C., Grigulis, K., Laurent, G., Lavorel, S., … Roumet, C. (2025). FAIRTraits: An enriched, FAIR-compliant database of plant traits from Mediterranean populations of 240 species. *Ecology*, *106*(9), e70219. https://doi.org/10.1002/ecy.70219

Alcántara, J. M., Verdú, M., Garrido, J. L., Montesinos-Navarro, A., Aizen, M. A., Alifriqui, M., Allen, D., Al-Namazi, A. A., Armas, C., Bastida, J. M., Bellido, T., Paterno, G. B., Briceño, H., Camargo de Oliveira, R. A., Campoy, J. G., Chaieb, G., Chu, C., Constantinou, E., **Delalandre, L.**, (…) Zamora, R. (2025). Key concepts and a world-wide look at plant recruitment networks. Biological Reviews, n/a(n/a). *https://doi.org/10.1111/brv.13177*

Delalandre, L., Violle, C., Coq, S., & Garnier, E. (2023). Trait–environment relationships depend on species life history. *Journal of Vegetation Science*, *34*(6), e13211. https://doi.org/10.1111/jvs.13211

Gaüzère, P., Blonder, B., Denelle, P., Fournier, B., Grenié, M., **Delalandre, L.**, Münkemüller, T., Munoz, F., Violle, C., & Thuiller, W. (2023). The functional trait distinctiveness of plant species is scale dependent. *Ecography*, *n/a*(n/a), e06504. https://doi.org/10.1111/ecog.06504

Munoz, F., Klausmeier, C. A., Gaüzère, P., Kandlikar, G., Litchman, E., Mouquet, N., Ostling, A., Thuiller, W., Algar, A. C., Auber, A., Cadotte, M. W., **Delalandre, L.**, Denelle, P., Enquist, B. J., Fortunel, C., Grenié, M., Loiseau, N., Mahaut, L., Maire, A., … Kraft, N. J. B. (2023). The ecological causes of functional distinctiveness in communities. *Ecology Letters*, *n/a*(n/a). https://doi.org/10.1111/ele.14265

Verdú, M., Garrido, J. L., Alcántara, J. M., Montesinos-Navarro, A., Aguilar, S., Aizen, M. A., Al-Namazi, A. A., Alifriqui, M., Allen, D., Anderson-Teixeira, K. J., Armas, C., Bastida, J. M., Bellido, T., Bonanomi, G., Paterno, G. B., Briceño, H., de Oliveira, R. A. C., Campoy, J. G., Chaieb, G., …, **Delalandre, L.** … Zamora, R. (2023). RecruitNet: A global database of plant recruitment networks. *Ecology*, *n/a*(n/a), e3923. https://doi.org/10.1002/ecy.3923

Delalandre, L., Gaüzère, P., Thuiller, W., Cadotte, M., Mouquet, N., Mouillot, D., Munoz, F., Denelle, P., Loiseau, N., Morin, X., & Violle, C. (2022). Functionally distinct tree species support long-term productivity in extreme environments. *Proceedings of the Royal Society B: Biological Sciences*, *289*(1967), 20211694. https://doi.org/10.1098/rspb.2021.1694

Delalandre, L., & Montesinos-Navarro, A. (2018). Can co-occurrence networks predict plant-plant interactions in a semi-arid gypsum community? *Perspectives in Plant Ecology, Evolution and Systematics*, *31*, 36–43. https://doi.org/10.1016/j.ppees.2018.01.001

Awards, honours and grants

2025: Two years of funding for my own research project, *The cultural evolution of gender inequalities*, through the Complexity Science Hub's Postdoc Program

2024: Open Science prize for Ph.D., **2500€** funded by the French ministry of Higher Education and Research

2024: Nomination by the editorial committee of the Journal of Vegetation Science, in the context of the award « More reproducibility in vegetation research » (see DOI: 10.1111/jvs.13224)

2020-2023: Specific Doctoral Scholarship for Students of the ENS de Lyon for my own research project on plant trait-environment relationships across scales.

Teaching experience

I taught ~130 hours in biology, ecology, evolution and statistics, for license to master students. I also obtained the « agrégation » French concours (for teaching biology and geology) in 2019.

Course	Language	Year	Academic level	Duration
Organism life cycles (UM)	French	2020	L1	18h
From genotype to phenotype (UM)	French	2020	L2	26h
Evolutionary ecology (UM)	French	2020 et 2021	L3	46h
Statistics (linear models) (UM)	French	2021	M1	12h
Functional ecology (UM)	French	2022	L2	14h
Functional diversity (UM)	French	2021 et 2022	M2	12h

UM: University of Montpellier.

Skills and qualifications

Languages:

- French (native), English (C2 Cambridge Advanced; fluent academic writing), Spanish (B2); German (beginner)

Computational & Technical:

- *Programming*: Advanced R (statistics, visualization), Python (data analysis), SQL, Bash
- *Modeling*: Agent-based models (NetLogo, R), GLMMs, Bayesian statistics, differential equations (Lotka-Volterra)
- *Tools*: GitHub (version control), LaTeX (manuscripts), Inkscape (figures), Zotero (references)
- *Data*: Multivariate analysis, null/permutation models, open-science workflows, database structuration

Research & Collaboration:

- Designed/led fieldwork and experimental studies in Ecology
- Mentored 3 interns in data collection and analysis
- Built and managed international trait databases (RECRUITNET, FAIRTraits)
- Long-term collaboartions with 4 international research teams

Other:

- European driving license (fieldwork logistics)

Mentoring

2022: Julie Tschambser, M1, *University of Montpellier* (four months)

2022: Lila Saugier, L3, *Ecole Normale Supérieure* (three months)

2022: Galadriel Fourtier, L2, *University of Montpellier* (two months)

Service

2020-2023: Organization and animation of the weekly scientific seminars of the Comparative Ecology team, *CEFE*, *Montpellier*.

Peer review

Functional ecology: four articles reviewed.

Scientific Reports (Springer Nature): two articles reviewed.

Ecology: one article reviewed.

Conferences

March 2025: Conference at CEFE on **project management and note-taking** with the Obsidian software.

June 2023: Seminar of the **CEFE Functional Ecology department**, on trait-environment relationships within species, between species, and between groups of species.

April 2023: Talk at **ECOVEG** (French seminar on plant ecology): empirical work on how plant life-history can affect trait-environment relationships.

March 2023: Invited talk at **ECOFOG** (French Guyana), on functional distinctiveness, its causes, properties, and effects on ecosystem productivity.

December 2021: Talk at the **British Ecological Society** symposium, on functional distinctiveness and forest productivity.

Workshops

May and October 2024: Interdisciplinary workshops on Science and Society (in French), *ENS*, *Lyon*

2023: Workshop on database management and Open Science (in French), *CEFE*, *Montpellier*

October 2022: Workshop on multilevel statistical modelling.

January 2020, October 2021, March 2022: International workshops on Functional Rarity in Ecology and Evolution, *CESAB*, *Montpellier*

Outreach

Contributed to writing YouTube science communication scripts for a French-speaking audience.

- Summarized evolutionary forces through a children's story. Associated a blog article.

- Explained basic concepts of cultural evolution in an accessible format, and discussed the parallels and differences with genetic evolution.

Referees

Eric Garnier, CNRS Researcher
PhD supervisor
CEFE, *Montpellier*, *France*
Email: eric.garnier@cefe.cnrs.fr

Cyrille Violle, CNRS Researcher
PhD supervisor
CEFE, *Montpellier*, *France*
Email: cyrille.violle@cefe.cnrs.fr

Alicia Montesinos-Navarro, CSIC Researcher
Master 2 internship supervisor
CIDE, *Valencia*, *Spain*
Email: ali.montesinos@gmail.com

Alex Mesoudi, Profesor of Cultural Evolution
University of Exeter, *UK*
Email: a.mesoudi@exeter.ac.uk